



Salimli Ayzek

Data Scientist. 1st-year master's student, SPbSTU, ICCS. Mathematics and Computer Science, Machine Learning and Artificial Intelligence.

thebloomsbridge - Work

GitHub

Hugging Face

ORCID

ResearchGate

eLibrary

Saint-Petersburg

Версия на русском

Download resume (ENG)

Download this portfolio

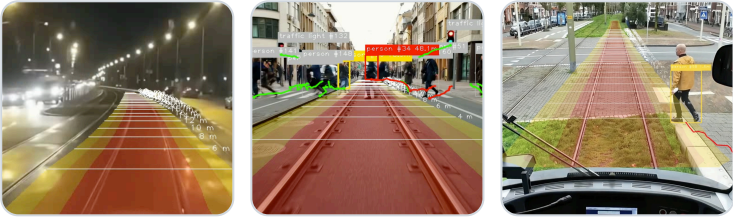
About

- Salimli Ayzek - master's student at SPbSTU (ICCS), majoring in Mathematics and Computer Science: Machine Learning and Artificial Intelligence. I have 1 year and 5 months of experience at The Blooms Bridge: AI systems integration (RAG, computer vision, etc.) and backend development. I primarily position myself as a Data Scientist and work with the ML cycle: data preprocessing and transformation, data analysis, feature selection, model training, hyperparameter tuning, result interpretation, data labeling, LLM training, model building (PyTorch, TensorFlow), deployment (LangChain + LangGraph), and computer vision tasks (YOLO, ResNet, MobileNet, ViT, U-Net, SegFormer, YOLO-seg, YOLO-pose, MediaPipe, OpenCV).
- My professional interests include computer vision, statistical and machine learning algorithms, neural networks, functional programming, optimization methods, statistics, and discrete mathematics.
- Outside work: travel, philosophy (especially how cognitive thinking and foundational ideas connect with computer science), football, and neue deutsche harte.

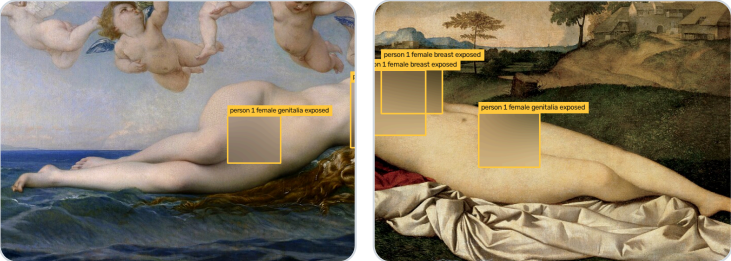
Projects

ML and AI

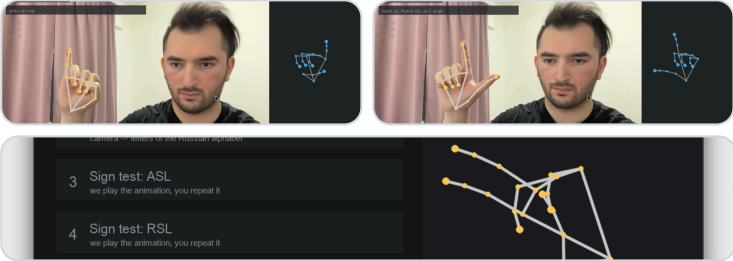
Object detection on dynamic RoI based on segmentation with tracking (weights available)



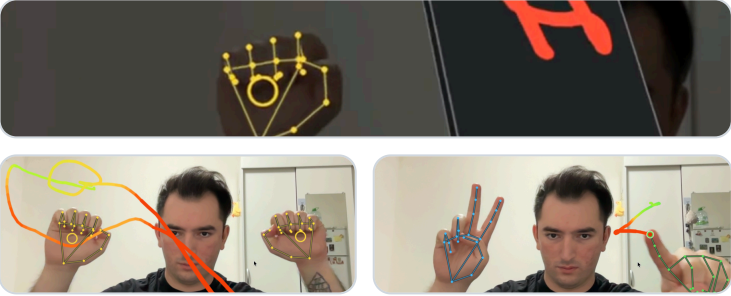
Real-time censorship API (for online meetings)



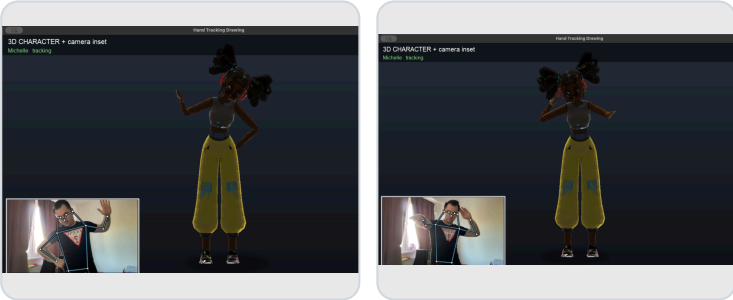
ASL, RSL sign languages to text (text to sign) translator



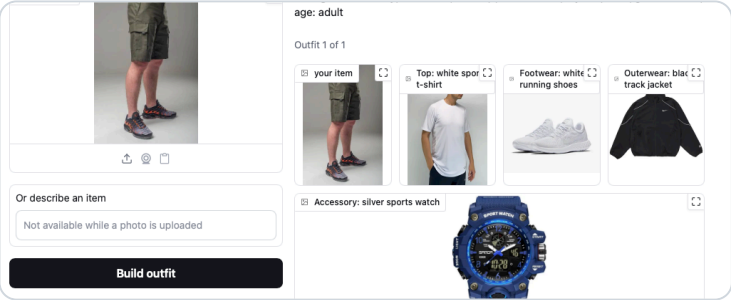
Hand tracking drawing (with 3D)



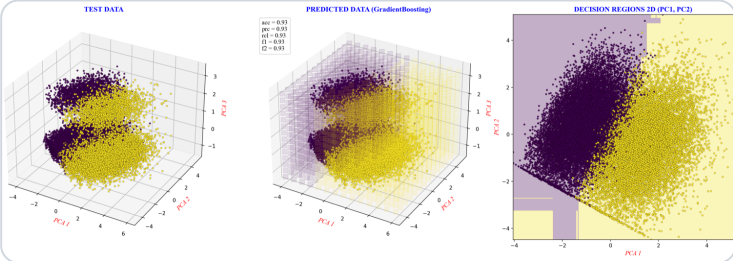
Pose driven 3D characters



Vision fashion recommender



Heart disease classifier

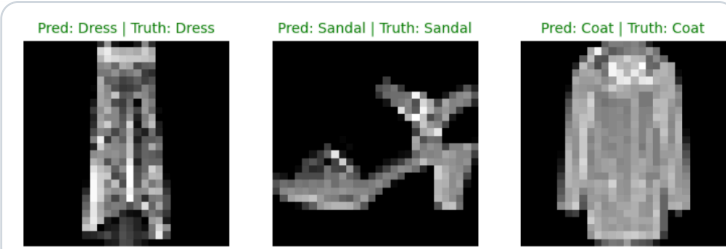


Qt MNIST recognizer



Fashion-MNIST classification with TinyVGG

Sarcasmic text classification (transformer NN)



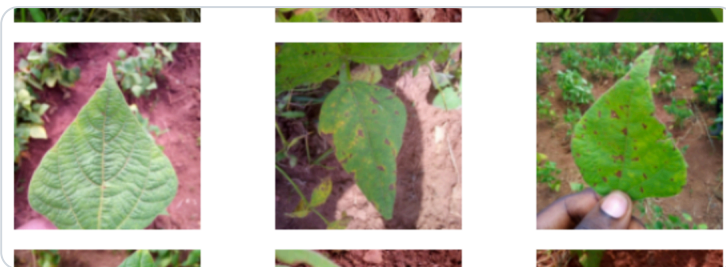
Confidence: 0.9799

Text: The guy found way to end with Global Hungry, but cant find his keys  
Prediction: Sarcasm  
Confidence: 0.9535

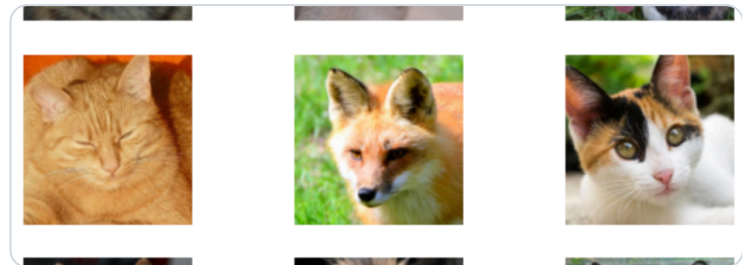
Text: Trump want's to write a law that allows Russians visit USA for free  
Prediction: Not Sarcasm  
Confidence: 0.9781

Text: Russia want's to improve young scientiests to do some experiments

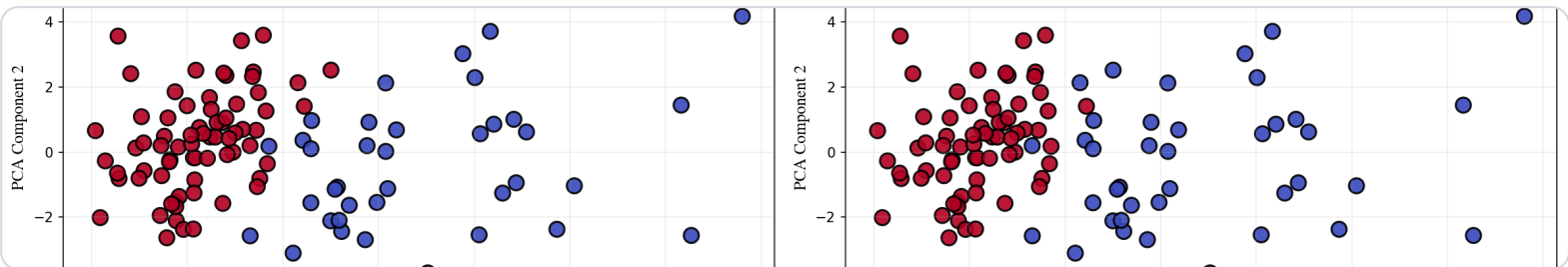
## Leafs health classification (GoogLeNet)



## Classification of cats, dogs and wild animals (CNN)

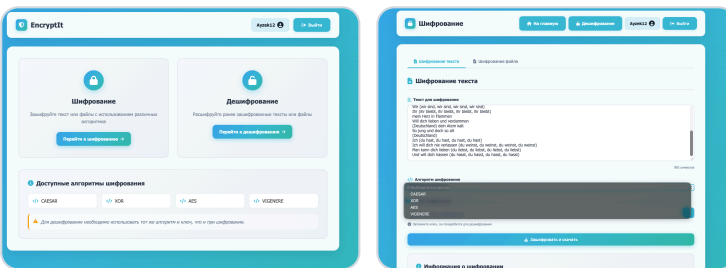


## Breast cancer classification

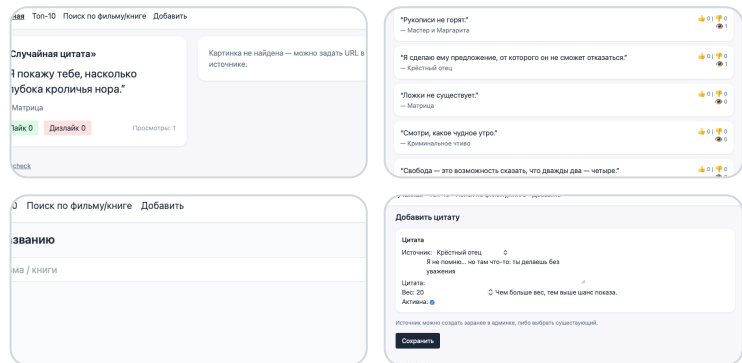


## Web

### Data encryption and decryption



### Quotes from movies and cartoons

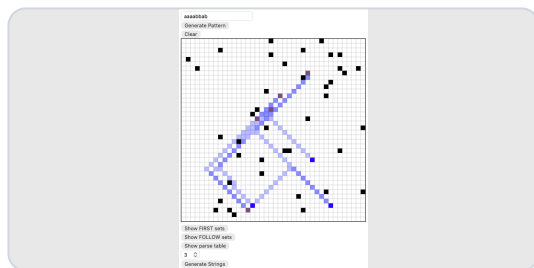


## Soft

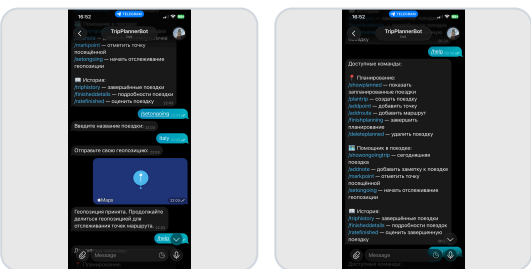
### Game of life on CA



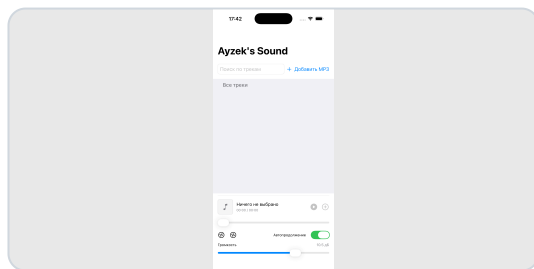
### LL(1) grammar with BFS semantics on Haskell



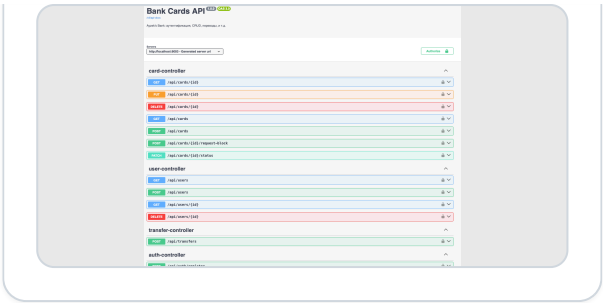
### Spring based TG-Bot: Trip planner



### AyzeksSound: Open source MP3 player for iOS



### Card management service



Math

MathLang presentations

легко описывает базовый принцип работы парсеров/ компиляторов/ анализаторов. Это только математическая модель, которая легко реализуется в коде.

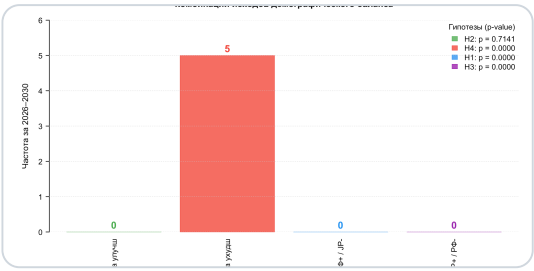
**Определение**

Формально это кортеж состоящий из:

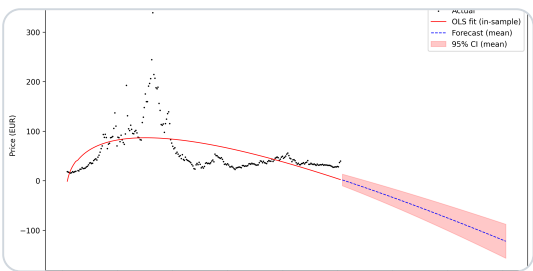
- Множество состояний -  $S$
- Множество входного и выходного алфавита -  $X, Y$
- Начальное состояние -  $s_0$
- Функция переходов -  $\delta : X \rightarrow S$
- Функция выходов -  $\lambda : X \rightarrow Y$

То есть DFA (Deterministic Finite Automaton):

Forecasting and comparison of demographic indicators of Russia and Japan



Regression analysis of the consequences of the reduction in gas export



Education collection (bachelor)



Education

**School**  
Complete secondary

Azerbaijan - Baku

**School №53**  
*Mathematics focus*

**Bachelor**

Russia - Saint Petersburg

**Peter the Great St. Petersburg Polytechnic University**  
*Institute of Computer Science and Cybersecurity*  
*Mathematics and Computer Science - AI Systems and Supercomputer Technologies*  
Graduate School of Artificial Intelligence Technologies

**Master's degree**  
1st year

Russia - Saint Petersburg

**Peter the Great St. Petersburg Polytechnic University**  
*Institute of Computer Science and Cybersecurity*  
*Mathematics and Computer Science - Machine Learning and Artificial Intelligence*  
Graduate School of Artificial Intelligence Technologies

Experience

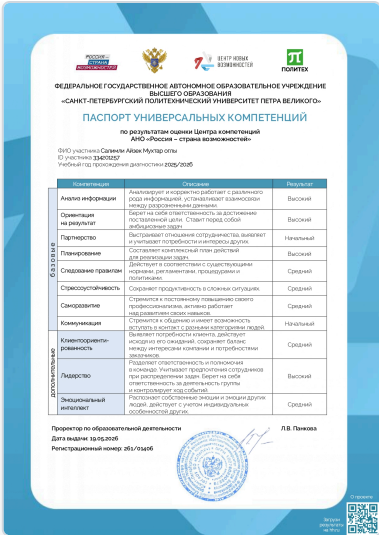
- [The Blooms Bridge](#) - ML Engineer (20.07.2024 - present).

Articles

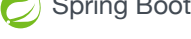
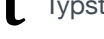
- [Construction of the Region of Interest based on segmentation<sup>1</sup>](#)
- [Computer vision for offside detection in football](#)

<sup>1</sup> - Recognized as one of the best articles.

Diplomas and Certificates



Skills

| Category                          | Type                        | Technologies / Libraries                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming Languages             | Multi-paradigm              |  C++  Python                                                                                                                                                                                                                                                                                                          |
|                                   | Object-Oriented (primarily) |  Java                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                   | Functional                  |  Haskell                                                                                                                                                                                                                                                                                                                                                                                                |
| Data science, ML, Computer vision | Data Manipulation           |                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                   | Data Visualization          |                                                                                                                                                                                                                                                                                                                   |
|                                   | Deep Learning (Frameworks)  |                                                                                                                                                                                                                              |
|                                   | Machine Learning            |                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                   | Scientific Computing        |                                                                                                                                                                                                                              |
|                                   | Computer Vision             |                                                                                                                                                                                                                                                                                                                                                                                                        |
| Quantum Computing                 | -                           |                                                                                                                                                                                                                                                                                                                                                                                                        |
| Frameworks                        | -                           |  FastAPI                                                                                                                                                                                                                                                                                                          |
| Databases (DBMS)                  | NoSQL                       |                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                   | SQL                         |  MySQL                                                                                                                                                                                                                                                                                                            |
| Mathematics                       | -                           | <ul style="list-style-type: none"><li>Discrete Math</li><li>Graph Theory</li><li>Linear Algebra</li><li>Calculus</li><li>Mathematical Statistics &amp; Probability Theory</li><li>Statistical Learning</li><li>Optimization Methods</li><li>Quantum Computing</li></ul>                                                                                                                                                                                                                     |
| Message Brokers                   | -                           |                                                                                                                                                                                                                                                                                                                                                                                                        |
| Containerization & Hosting        | -                           |  Docker  Kubernetes  Firebase  Render                                                                                                   |
| CI/CD & Cloud                     | -                           |  Jenkins  Terraform  Ansible  HEAT OpenStack Heat  |
| Documentation                     | Documentation & Reports     |  LaTeX  Markdown  Typst                                                                                                                                                                                                      |
|                                   | Presentations               |                                                                                                                                                                                                                                                                                                                                                                                                        |

Learning Hub (on progress)

- [Computer Science](#)
- [Mathematics](#)

# Contacts

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- [Telegram](#)
- [GitHub profile](#)
- [Gitea profile](#)